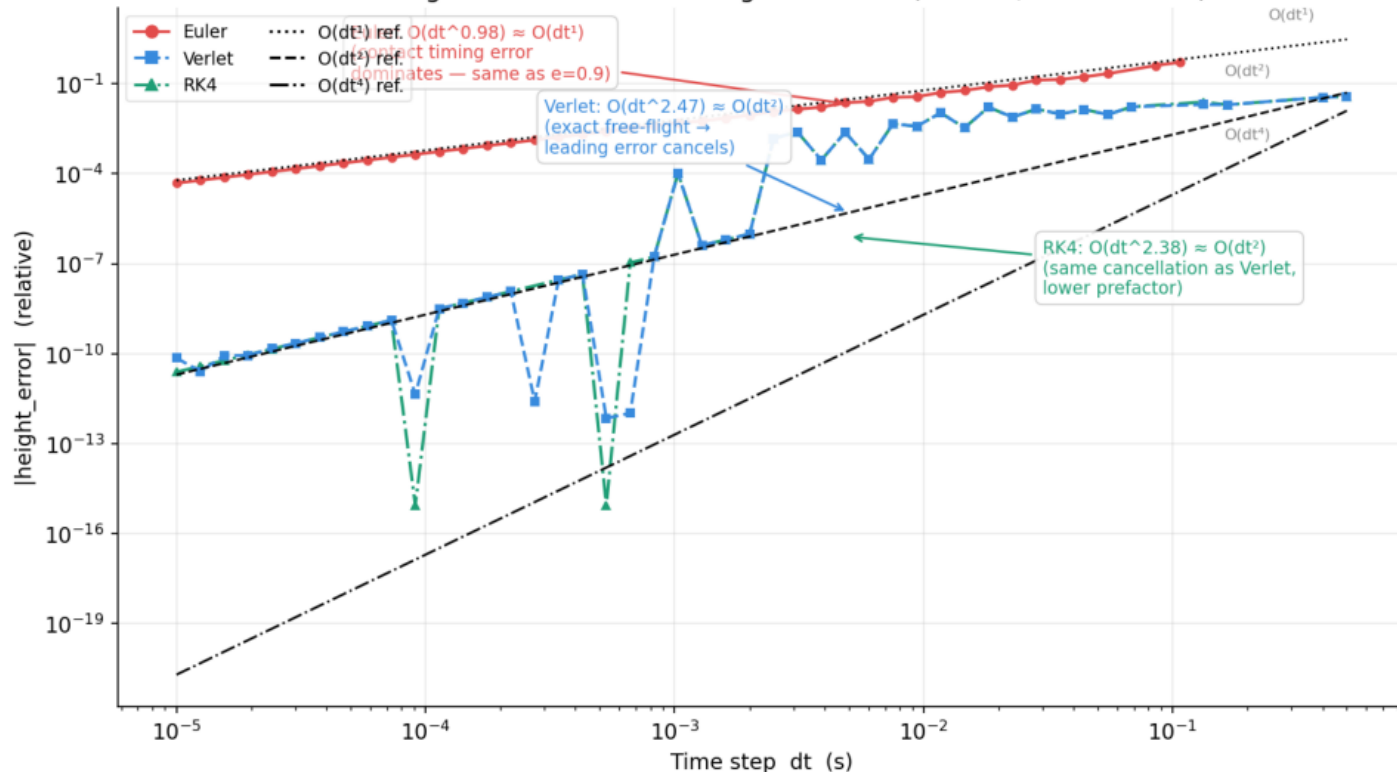


Peak height error vs dt — convergence order (e = 1.0, conservative)



CONVERGENCE ORDER (e = 1.0)

Measured slopes (log-log fit):

Euler → $O(dt^{0.98})$ $R^2=1.000$

Verlet → $O(dt^{2.47})$ $R^2=0.830$

RK4 → $O(dt^{2.38})$ $R^2=0.918$

Unlike e=0.9, integrators are NOW distinguishable!

Why Euler stays $O(dt^1)$:

First-order position error adds an $O(dt)$ term that does NOT cancel → same contact timing floor as before.

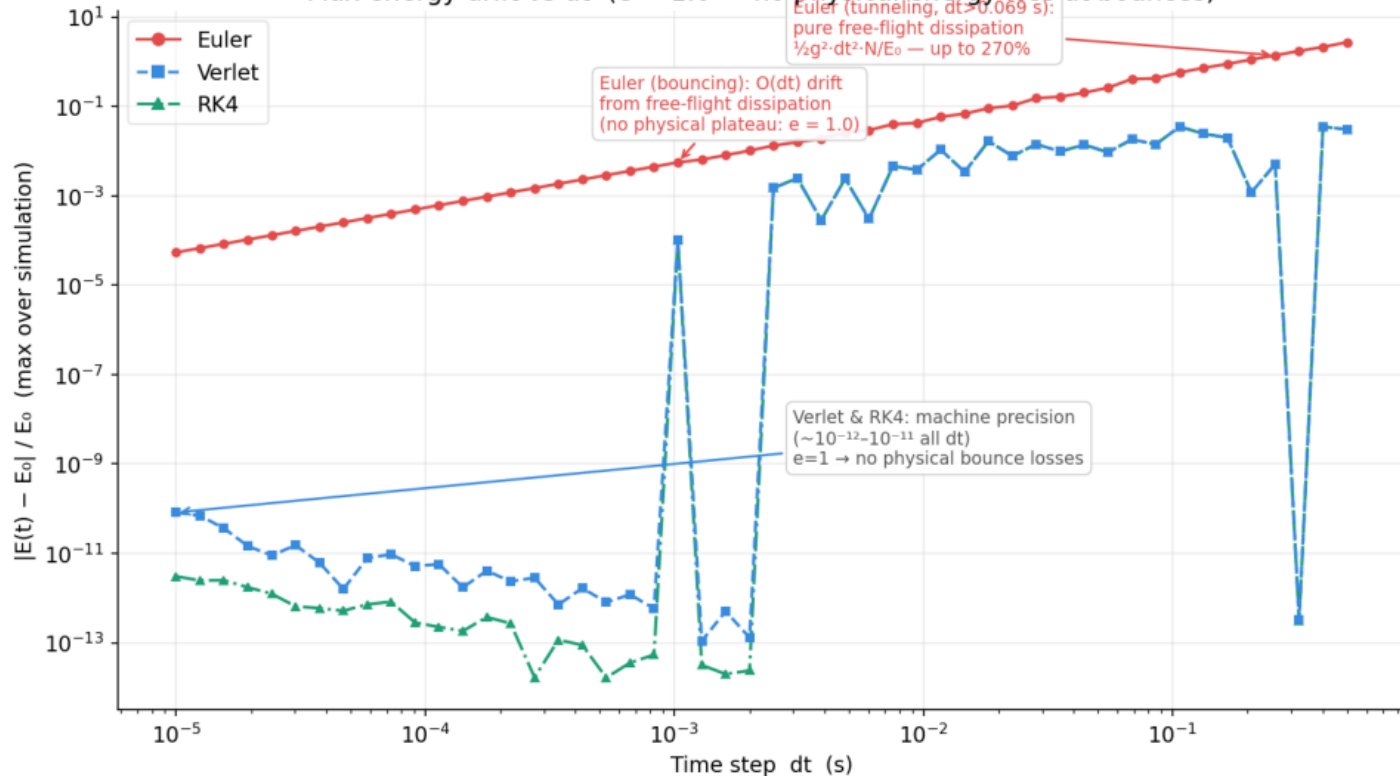
Why Verlet/RK4 show $O(dt^2)$:

Exact free-flight integration → position at contact is exact → the penetration depth and the extra velocity exactly cancel in the peak-height formula, leaving only $O(dt^2)$ residual.

Why NOT $O(dt^4)$ for RK4:

Both Verlet and RK4 are EXACT for quadratic trajectories, so the cancellation is the same for both → $O(dt^2)$ for both. RK4's higher order only matters for non-quadratic ODE.

Max energy drift vs dt (e = 1.0 — no physical energy loss at bounces)



ENERGY CONSERVATION (e = 1.0)

With $e = 1$, there is NO physical energy loss at bounces. All drift is NUMERICAL.

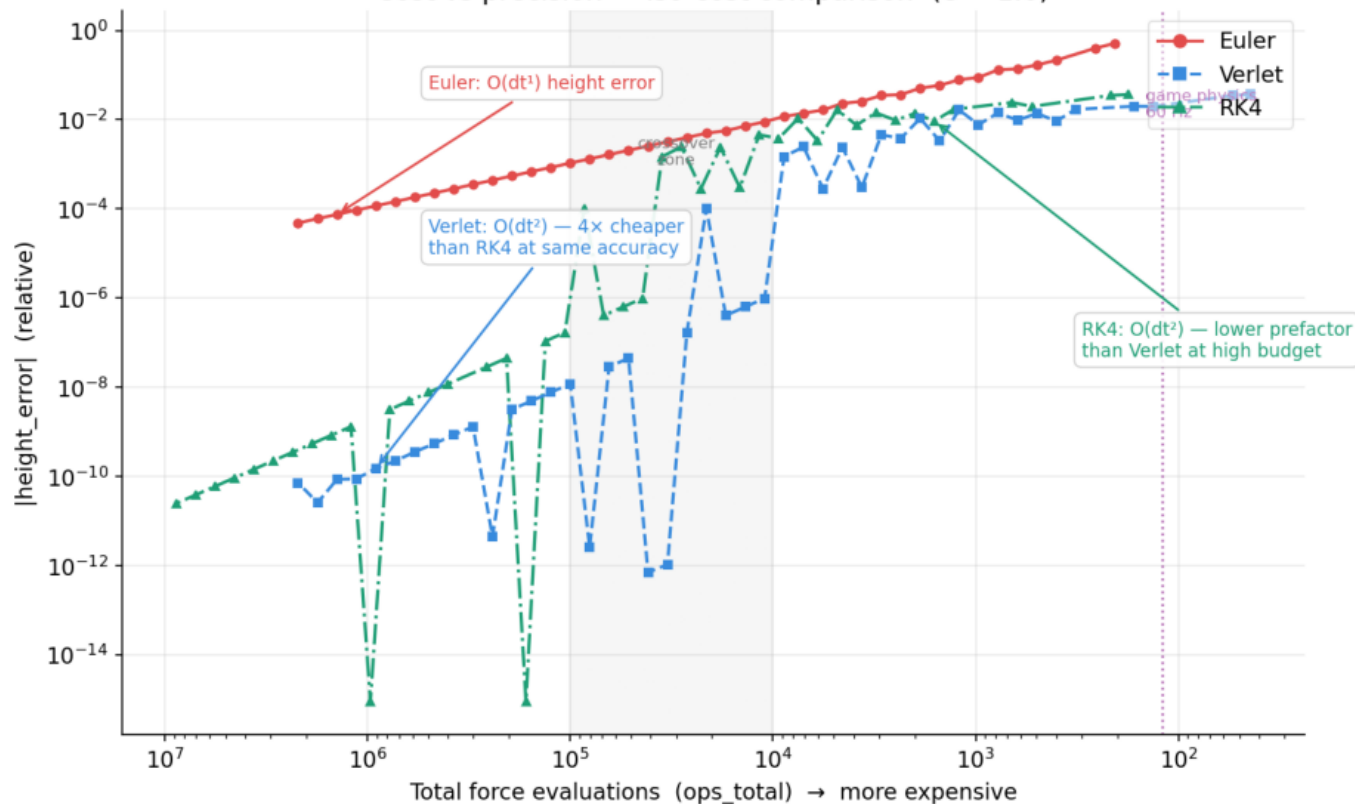
Verlet & RK4 (bouncing regime):
Machine precision $\sim 10^{-12}$ for all dt where bounces occur.
Free-flight exact + elastic bounce $\rightarrow$ near-zero energy error.

Euler (bouncing regime):
 $O(dt)$ drift from $\frac{1}{2}g^2 dt^2 / \text{step}$ dissipation in free flight.
No plateau: with $e=1$ there is nothing to mask the integrator error at small dt .

Large dt — tunneling:
Euler: $dt > 0.069$ s, drift up to 270% (pure dissipation).
Verlet/RK4: $dt > 0.085$ s, near-zero (no drift, no loss).

$\rightarrow$ Energy conservation gap between Euler and Verlet/RK4 is now fully unmasked: $\sim 10^8 \times$ at fine dt .

Cost vs precision — iso-cost comparison (e = 1.0)



COST vs PRECISION (e = 1.0)

X-axis: total force evaluations
(Euler=1/step, Verlet=1/step, RK4=4/step) ← left = costlier

With e=1, all three integrate
NOW show distinct orders:

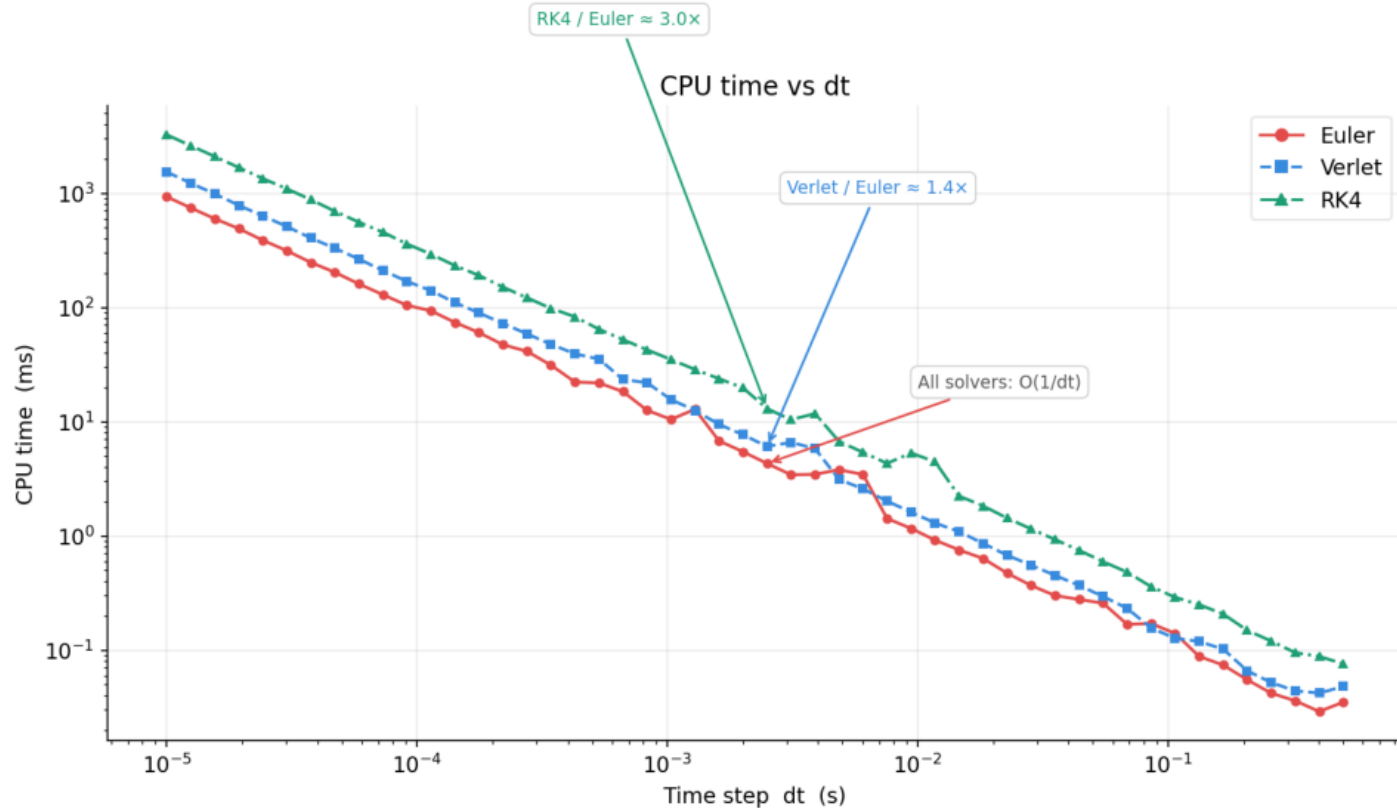
Euler → $O(dt^1)$ height error
Verlet → $O(dt^2)$ — same cost/step
RK4 → $O(dt^2)$ — 4x costlier

Verlet is the dominant choice:

Same $O(dt^2)$ order as RK4.
Same cost/step as Euler.
→ Always better than Euler
→ Always cheaper than RK4
at the same accuracy.

RK4 lower prefactor (error ~3x
lower at same dt), but costs 4x
more → Verlet wins on iso-cost.

For conservative physics,
Verlet is uniquely optimal: $O(dt^2)$ accuracy at $O(dt^1)$ cost.



CPU TIME

All solvers scale as $O(1/dt)$:
halving dt doubles CPU time.

Measured cost ratios vs Euler
at median dt:

Verlet : $\sim 1.41\times$

RK4 : $\sim 3.02\times$

With $e=1.0$, the accuracy gap
between integrators is visible
in height error (Plot 1).

Cost-accuracy trade-off:

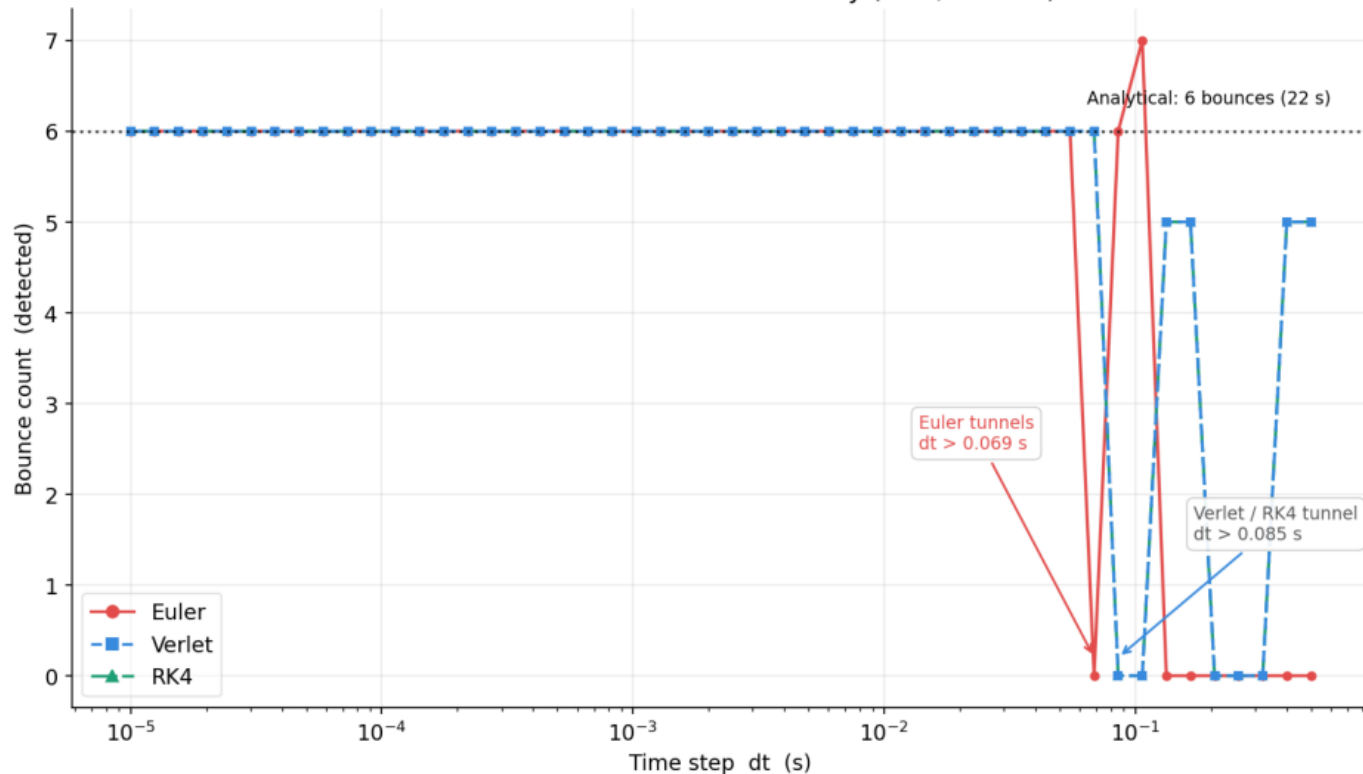
Euler : $O(dt^1)$, cheapest

Verlet : $O(dt^2)$, $1.4\times$ Euler

RK4 : $O(dt^2)$, $3.0\times$ Euler

→ Verlet is uniquely optimal:
 $O(dt^2)$ accuracy at $O(dt^1)$
cost/step – the best achievable
for this contact model.

Bounce count vs dt — solver stability (22 s, e = 1.0)



BOUNCE COUNT STABILITY

Analytical: 6 bounces in 22 s.
With $e=1$, every bounce returns to $z_0=20$ m $\rightarrow$ constant flight time $T = 2v/g \approx 3.83$ s.

Zone 1 – Tunneling (large dt):

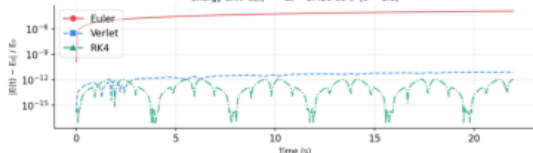
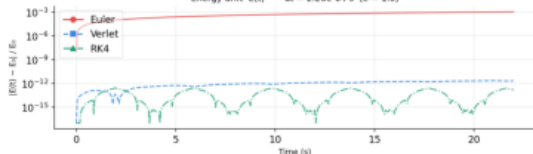
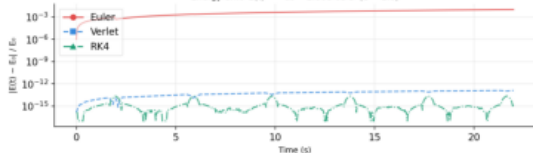
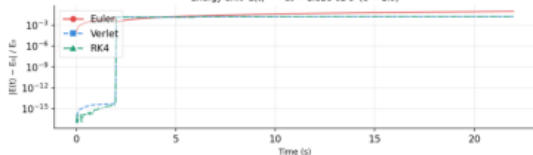
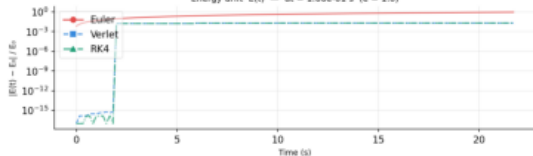
Euler : $dt > 0.069$ s $\rightarrow$ 0 bounces
Verlet/RK4: $dt > 0.085$ s
With $e=1$, approach velocity is constant across all bounces $\rightarrow$ threshold doesn't degrade.

Zone 2 – Stable (small dt):

All solvers: 6 bounces.
Verlet/RK4 show 5 bounces for 4 dt values (near tunneling threshold – 1 bounce missed).

Zone 3 – Euler transition:

Unstable (1–5 bounces) near tunneling threshold.
Verlet/RK4: sharper transition.

Energy drift $E(t)$ — $\Delta t = 2.42\text{e-}05$ s ($e = 1.0$)Energy drift $E(t)$ — $\Delta t = 2.20\text{e-}04$ s ($e = 1.0$)Energy drift $E(t)$ — $\Delta t = 2.00\text{e-}03$ s ($e = 1.0$)Energy drift $E(t)$ — $\Delta t = 1.82\text{e-}02$ s ($e = 1.0$)Energy drift $E(t)$ — $\Delta t = 1.66\text{e-}01$ s ($e = 1.0$)ENERGY DRIFT $E(t)$ ($e = 1.0$)

Each panel shows $|E(t) - E_0| / E_0$ over the full simulation.
 E_0 = initial kinetic + potential.

NO staircase! With $e=1.0$:
 Each bounce is perfectly elastic
 — no physical energy loss.
 Any drift is purely numerical.

Verlet / RK4:
 Flat at machine precision
 ($\sim 10^{-17}$) for all Δt .
 Exact free-flight + elastic
 bounce = near-zero error.

Euler:
 Steady upward slope from
 $-\log(\Delta t)^2$ dissipation per step.
 Slope steeper at larger Δt .
 No plateau masking it.

Coarsest Δt (1.66e-01 s):
 Ball tunnels — no bounces.
 Euler: pure linear drift.
 Verlet/RK4: machine ϵ .

The energy gap between Euler
 and Verlet/RK4 is now $\sim 10^{-6}$
 at fine Δt — fully unmasked.